INCH-POUND
MIL-DTL-24643/9E
22 August 2002
SUPERSEDING
MIL-C-24643/9D
22 November 1994

DETAIL SPECIFICATION SHEET

CORD, ELECTRICAL, TYPE LSMRI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor - class K stranding, uncoated (see table I for size).
Second - Ethylene propylene rubber or cross-linked polyethylene insulation. Standard identification code shall be applied by method 3.
Third - The specified number of conductors cabled with a lay not greater than that specified (see table I).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:56Z
Check the source to verify that this is the current version before use.

Table I. Details

Military part no. M24643/9	Type and size	Conductors		Diameter over insulation		Lay of conductors maximum (inch)	Conductor resistance 1000 feet (max ohms)
		No.	AWG	Minimum (inch)	Maximum (inch)		
-01UN	LSMRI-D-1	2	20(class K)	0.068	0.075	2	11.05
-02UN	LSMRI-D-2-1/2	2	16(class K)	0.092	0.102	3	4.36
-03UN	LSMRI-T-2-1/2	3	16(class K)	0.092	0.102	3	4.36

EXAMINATION AND TESTS:

Basic electrical: Requirements:

- Conductor resistance - ohms/1000 feet at 25°C, maximum ..... (see table I)
- Voltage withstand - conductor to conductor, volts, root mean square, minimum
  - AWG 20 Class K 1000
  - AWG 16 Class K 1500
- Insulation resistance - megohms -1000 feet, minimum 100
- Conductor continuity ..... No failure

Group A:

- Visual and dimensional..... No failure

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50
  - Jacket (When tested at 200°C)..... Not applicable

Physicals (unaged)

- Insulation
  - Ethylene propylene rubber
    - Tensile strength - lb/in^2, minimum 700
    - Elongation - percent, minimum 250
  - Cross-linked polyethylene
    - Tensile strength - lb/in^2, minimum 1800
    - Elongation - percent, minimum 250

Group C:

Physicals (aged)

- Insulation
  - Ethylene propylene
    - Air oven
      - Tensile strength - percent of unaged, minimum 75
      - Elongation - percent of unaged minimum 75
  - Cross-linked polyethylene
    - Air oven
      - Tensile strength - percent of unaged, minimum 80
      - Elongation - percent of unaged minimum 80
- Heat distortion - percent of unaged, maximum 30

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:56Z
Check the source to verify that this is the current version before use.

Group D:

Flame propagation (cable) ..... Not Applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following

	Requirements:
Acid gas equivalent (insulation) - percent, maximum	18
Smoke index, (insulation) maximum	45
Toxicity index, (insulation) maximum	1.5

UNIT ORDERING LENGTH:

Type	Size	Feet (nominal)
LSMRI-D-	1	600
LSMRI-D-	2-1/2	600
LSMRI-T-	2-1/2	500

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-007)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:56Z
Check the source to verify that this is the current version before use.